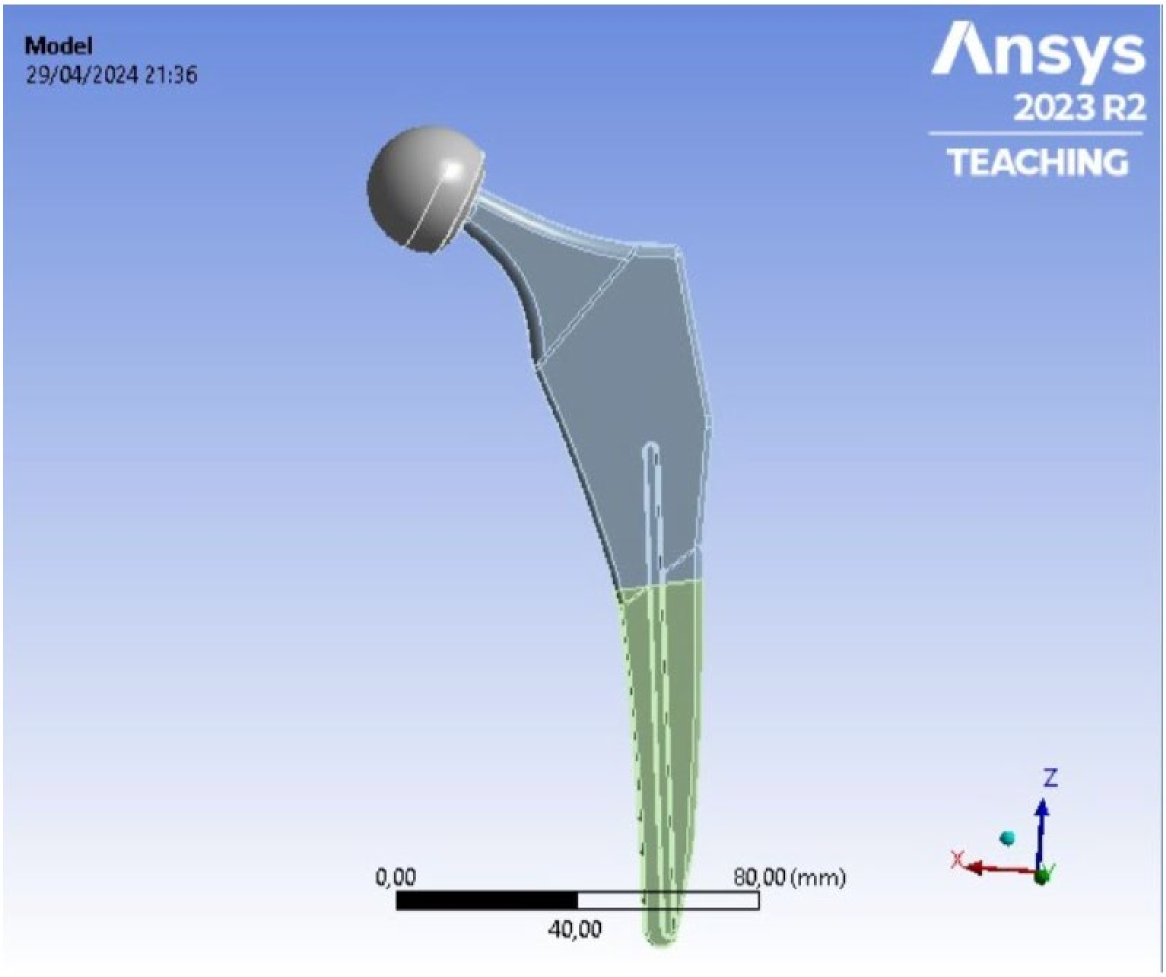
Hip Implant FEA Analysis



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1 Introduction

1.1 Design problem

The hip joint, crucial for mobility, is susceptible to damage from ageing and physical activities, which causes pain and often requires surgical intervention in the form of hip implants. Although, they are designed to last for several years, minimizing the need for additional surgeries, hip implants may experience failures such as loosening or internal deterioration over time, so detecting and monitoring implant health is vital. This can be achieved though approaches like vibration-based non-destructive testing (NDT), which relies on analysing multiple natural frequencies within a specific range to assess structural integrity.

This report delves on assessing the natural frequencies and fatigue life of four hip implants with different geometries through FEA analysis to check whether the designs met the following requirements:

- Lightweight
- Natural frequencies between: 200Hz-3000Hz
- Effective life of at least 15 years

2 Methods

2.1 Materials

The chosen Hip implant is to be tested with two different materials: Titanium alloy Ti-6AL-4V and Stainless Steel 316L. Both materials were found within the contents of material universe on Ansys, and the S-N curves were imported from CES Edupack.

Material	Youngs Modulus (GPa)	Tensile Strength Ultimate (MPa)	Density(kgm^-3)
Titanium alloy Ti-6Al-4V	1145	920	4430
Stainless Steel 316L	1975	522	7970

Table 1: Material Properties from Ansys

Stainless Steel and titanium are commonly used for hip implants due to their excellent mechanical properties, biocompatibility, and corrosion resistance. Stainless Steel, especially 316L, is strong and resists corrosion, ensuring durability and safety in the body. Titanium alloys like Ti-6Al-4V offer a high strength-to-weight ratio, superior corrosion resistance, and the ability to bond with bone tissue through osseointegration, which enhances implant stability and longevity. [2]

2.2 Boundary Conditions

For the simulation the stem of the implant is fixed at roughly halfway down to simulate the connection between the implant and femur bone. This constrain is crucial for the vibration analysis where the status of the support is assed to determine if it's been broken or damaged.

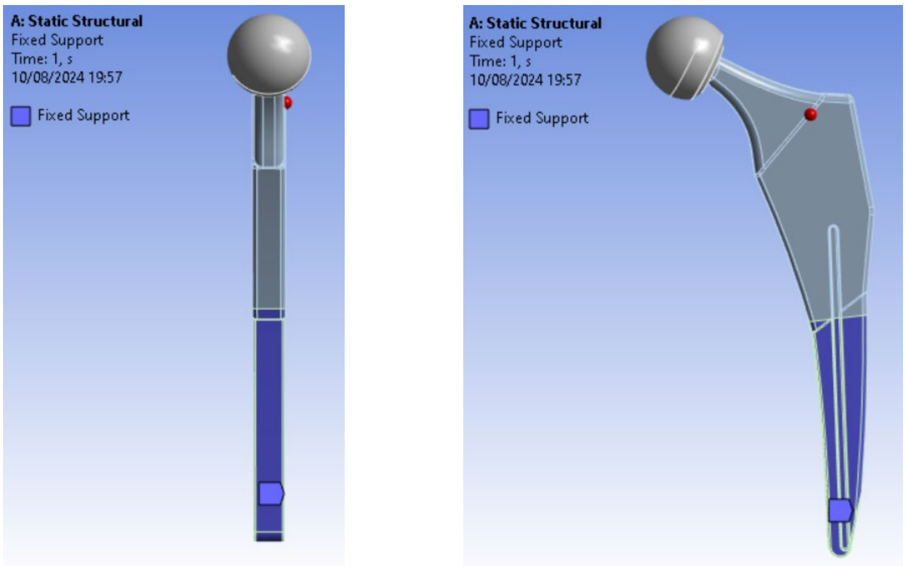


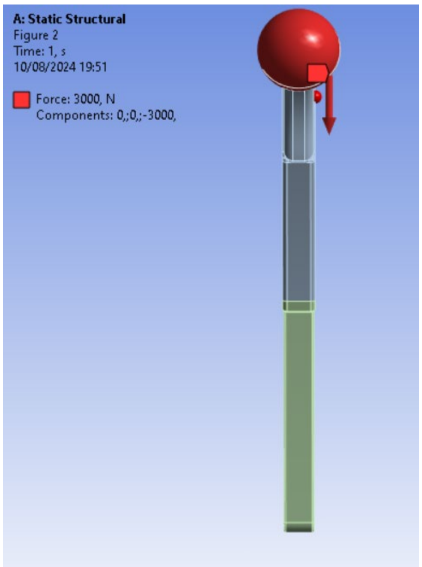
Figure 1 & 2: Fixed support boundary condition halfway down the stem

The fixed support restricts movement in all directions around the lower half of the stem, setting all translations and rotations at the support node to zero, resulting in zero degrees of freedom.

There would likely be some movement between the bone and implant, and other biological features such as flesh and bone marrow influence the fixation of other parts of the implant. Further analysis of these factor would help reduce modelling errors.

2.3 Loads

The implant is also subjected to a vertical force (along the axis of the body) which varies between 0 to 3000N. This intends to simulate the normal forces of walking and performing other daily activities. Additionally, standard earth gravity is also placed acting along the Z-axis to simulate all forces acting on the implant in real life, but this will have a small influence on the results.



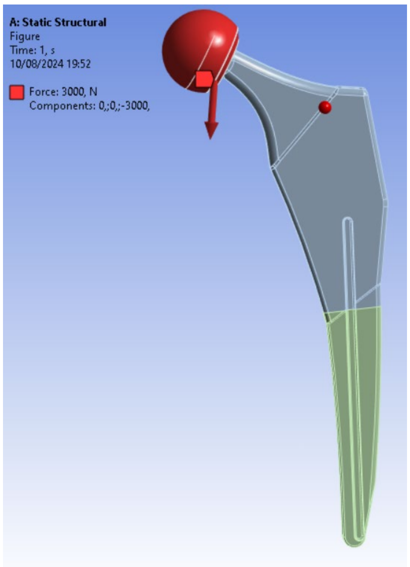


Figure 3 & 4: Load acting on head of implant in the Z- direction

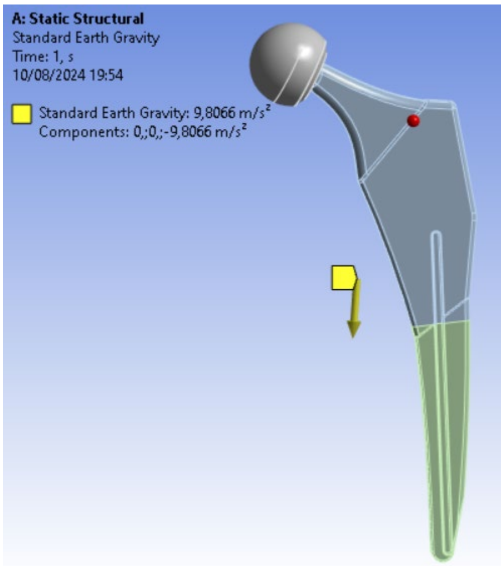


Figure 5: Applied standard earth gravity

Although the angle of the load would shift and change during different activities, here the load has been applied to all the surface of the head as a simplified model. This could be a potentially modelling error.

2.4 Natural Frequency Study Parameters

A natural frequency is the rate at which an object vibrates when it is disturbed from its rest position and allowed to oscillate freely. This is important because it helps ensure the implant's structural integrity and durability under physiological loads. Knowing the natural frequencies allows engineers to design implants that avoid resonant vibrations, which could lead to fatigue failure or loosening of the implant over time, thus enhancing the implant's longevity and reliability.

In this simulation, the aim is to achieve 5 natural frequencies between 200Hz-3000Hz. During the simulation the following boundary conditions are applied:

- Fixed support into femur
- No external force or mass

Using the following equation [3]

$$w = \sqrt{\frac{k}{m}}$$

demonstrates that increasing the stiffness (k) or decreasing the mass (m) will increase the natural frequency.

2.5 Fatigue Study Parameters

Fatigue analysis involves testing an object through numerous load cycles to determine if it will fail over its lifetime. As objects endure forces and stress, microscopic imperfections in the material can expand into larger cracks, ultimately causing failure.

The aim of this simulation is to determine if its effective life will be of at least 15 years (equivalent to 5 million loading cycles). As this simulation aims to mimic user's various activities, the load will fluctuate between zero and 3000N (max load), so the loading type is set to zero based.

Additionally, a mean stress concentration curve is applied, as the mean stress will be greater than zero. As both Titanium and Stainless Steel are ductile materials, the Gerber mean stress method is used to accurately assess fatigue life under varying mean stress conditions.

Using the following equation [4]

$$S_{ca} = \frac{S}{1 - \left(\frac{\sigma_{mean}}{S_u}\right)^2}$$

the impact of mean stress on fatigue life assuming a parabolic relationship instead of a linear one can be analysed.

By using the Fatigue tool's life feature uses the S-N curve to estimate the number of cycles an object can endure.

Furthermore, Von Mises stress is selected due to the material's ductility. This typically results in a lower estimated fatigue life compared to using principal stress, as Von Mises stress considers the combined effects of all stress components (including shear stress).

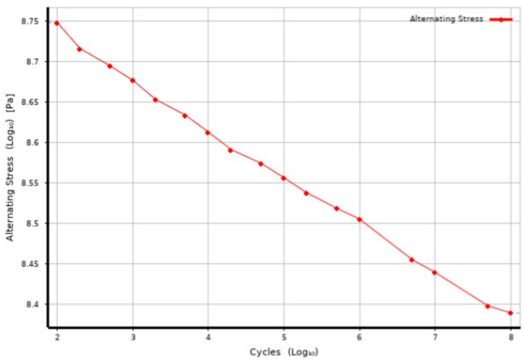


Figure 6: Stainless Steel 316L S-N curve

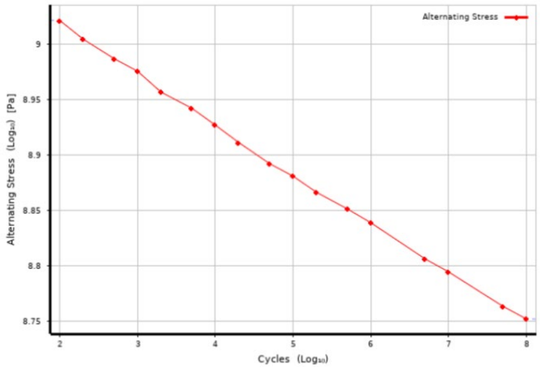


Figure 7: Titanium Ti-6Al-4V S-N curve

Details of "Fatigue Tool"

Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Zero-Based
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	Gerber
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

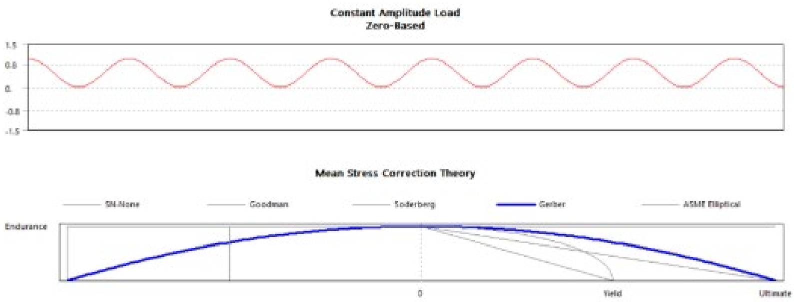


Figure 8 & 9: Fatigue tool settings

3 Results

3.1 Mesh Refinement

A mesh is a network of small, interconnected elements that divides an object's geometry. To obtain a high-quality mesh that computes efficiently, a manual mesh convergence study was conducted. This involved progressively decreasing the global element size until the maximum resulting stress stabilized, meaning that further reductions in element size had negligible effects on the stress.

Global Element Size (mm)	%Elements with Aspect ratio <3	%Elements with aspect Ratio >10	Minimum Jacobian Ratio	Maximum Stress (MPa)
5.0	99.85	0.05	0.09	305.63
4.0	99.26	0.04	0.32	291.15
3.0	100	0.023	0.08	312.68
2.0	99.71	0.031	0.08	326.45
1.5	99.98	0.025	0.08	323.9
1.25	99.96	0.025	0.08	330.13
1.0	99.98	0.025	0.085	331.99
0.8	99.99	0.023	0.09	347.88
0.5	99.76	0.028	0.085	314.78

Table 2: Recording for mesh convergence

Although analysis of the results indicated that perfect convergence had not been achieved, and further mesh refinement could improve accuracy, meshing exceeded one hour at this point. Additionally, the stress output changed by only 7% between the 2 mm and 0.8 mm studies, and by 1% between the 1.25 mm and 1 mm studies. Therefore, the mesh was considered sufficiently refined to provide a good level of accuracy while maintaining reasonable computation time. As a result, a global element size of 1.25 mm was selected.

3.2 Element Quality Metrics

Two additional critical factors to consider were aspect ratios (which indicate how close an element is to an ideal shape) and Jacobian ratios (which measure element distortion).

Although the ideal aspect ratio is 1, due to the decrease in accuracy of integral as ratio increases, the study aimed to increase the percentage of elements with aspect ratio below 3.

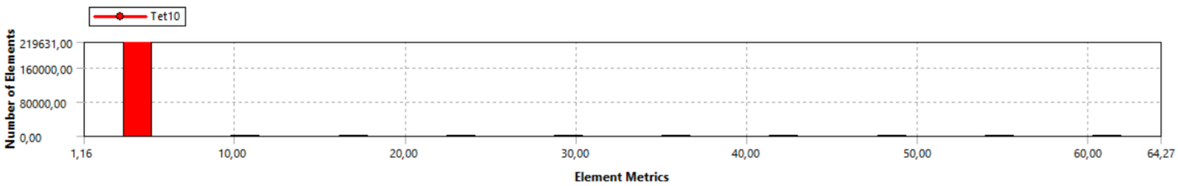


Figure 10: number of elements VS Aspect ratio

The chart indicates that most elements have an aspect ratio of 4.32, and only a small percentage were larger than 5, suggesting a good overall aspect ratio.

Additionally, the ideal Jacobian ratio is 1 and Ansys recommends that all elements should have a Jacobian greater than 0, but a useful rule of thumb is that all elements should have a Jacobian ratio below 40.

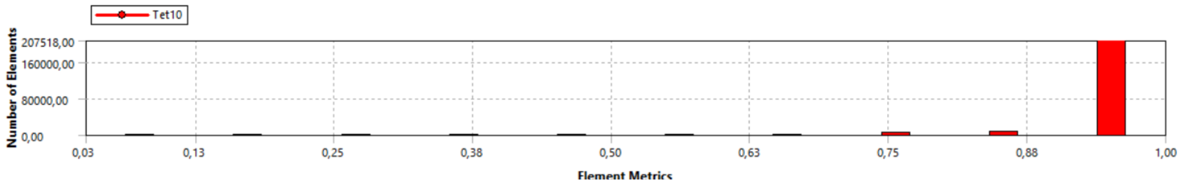


Figure 11: number of elements VS Jacobian ratio (Corner Nodes)

The chart indicates that most elements are 0.951, just below 1, with the smallest having a ratio of 0.08, suggesting minimal distortion of elements.

3.3 Sanity Checks

To ensure the studies were set up correctly, and that the results were sensible and accurate, sanity checks were undertaken.

All tests were successful, confirming that the boundary conditions were correctly set up and that the design's shape and geometry were sufficient to pass static analysis.

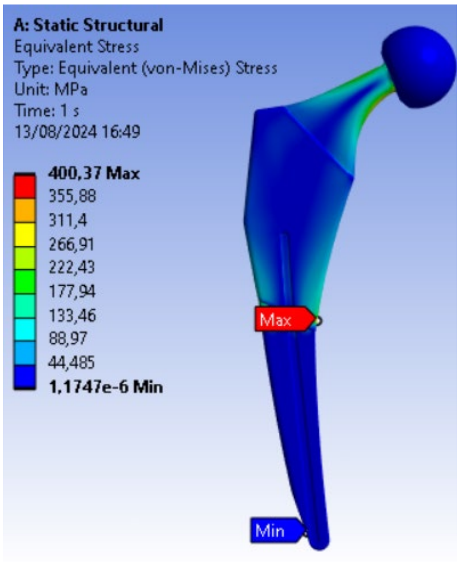


Figure 12: Titanium Max stress

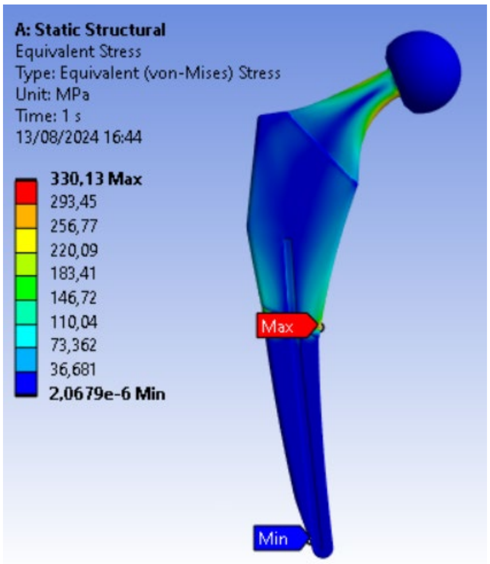


Figure 13: Stainless Steel Max stress

3.4 Simulation Results – Original Design

3.4.1 Vibrations Study

Titanium Alloy

Figure 14: Titanium Alloy frequency results

Stainless Steel

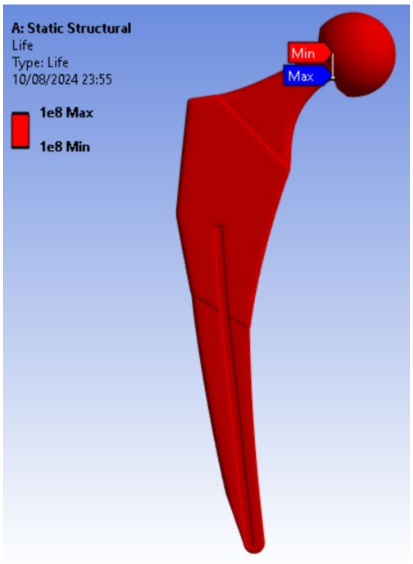
Figure 15: Stainless Steel frequency results

In both cases, there were 3 vibrational model between 200Hz-3000Hz, but not the desired 5.

3.4.2 Fatigue Study

Both material implants successfully exceeded the target of 15 million load cycles before failure, with the maximum alternating stress remaining below the fatigue limit. This indicates, in theory, that the implants could endure an infinite number of cycles.

Titanium Alloy **Stainless Steel**



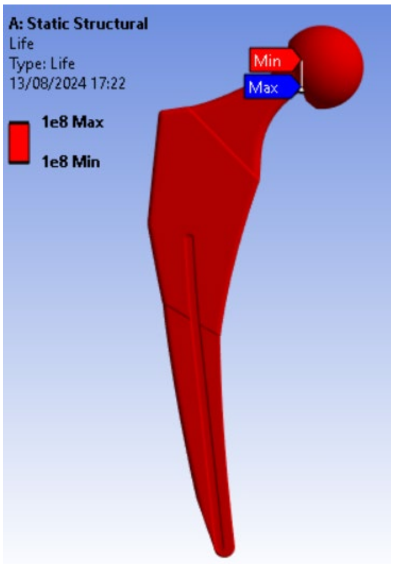


Figure 16: Fatigue life results for Titanium Alloy *Figure 17: Fatigue life results for Stainless Steel*

3.5 Rationale for Improvement

The aim of the redesign is to ensure all 5 of the vibrational modes are between 200Hz-3000Hz. As previously noted, from the natural frequency equation, increasing the mass and reducing the stiffness of the implant will result in lower vibrational modes, potentially increasing the number within the desired range.

Design requirements:

- **Increase Mass:** Increase thickness of certain parts of the implant, like the neck. However, ensuring that the changes maintain proper fit and do not interfere with functionality is key.
- **Decrease Stiffness:** Incorporate hollow structures on the outside of the implant to create more flexible zones. Although these significantly reduce stiffness, they also decrease mass, creating a slight opposing effect.

Figure 18: First design iteration

Figure 19: Increased thickness of neck

Further static analysis was carried out to determine if the design would stand the maximum stress and if it could still be used.

Material	N. elements	Max stress (von Mises) (MPa)	Yield stress (MPa)	Static failure (pass/fail)
Titanium alloy Ti-6Al-4V	166611	400	786-910	pass
Stainless Steel 316L	´	330	170 – 310	fail

Table 3: Static analysis on altered design

While Titanium passed the static failure analysis, stainless steel did not, as the maximum stress was higher than the yield, which indicates that the hip implant would no longer be suitable for use.

3.6 Simulation Results – Improved Design

3.6.1 Vibrations Study

Titanium Alloy

Figure 20: Titanium Alloy frequency results first iteration

Stainless Steel

Figure 21: Stainless Steel frequency results first iteration

The first iteration resulted in still 3 vibrational modes between 200Hz-3000Hz for both materials. Although a decrease was noted in the frequency which proved that the redesign had its desired effect, further alterations need to be made to get the desired 5 modes within the desired range.

3.6.2 Fatigue Study

Again, both material implants sustain the expected lifetime of a million cycles without significant internal damage.

Titanium Alloy **Stainless Steel**

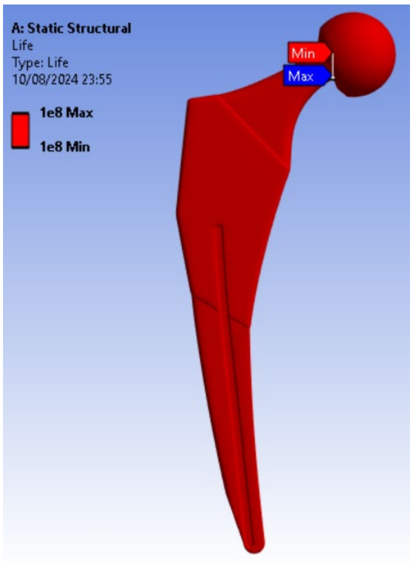


Figure 22: Fatigue life results for Titanium Alloy

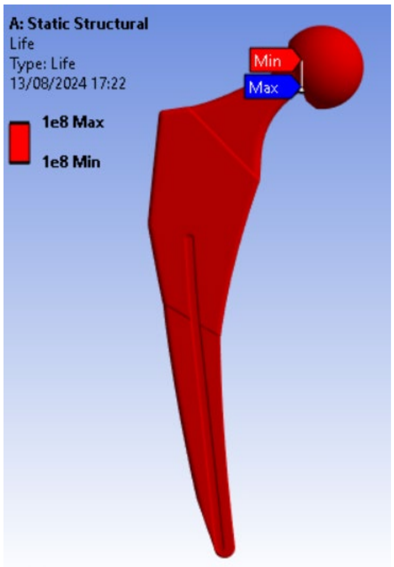


Figure 23: Fatigue life results for Stainless Steel

3.7 Further Iteration

Since the previous iteration didn't improve the vibrational modes as desired, further changes were made to the design to further decrease its stiffness. More prominent cut out channels were added to the sides.

Figure 24: Second design iteration

3.7.1 Vibrations Study

Titanium Alloy

Figure 25: Titanium Alloy frequency results second iteration

Stainless Steel

Figure 26: Stainless Steel frequency results second iteration

This second iteration resulted in 4 vibrational modes for stainless steel and 3 vibrational modes for titanium between 200Hz-3000Hz. Although it is closer to the desired range, further iterations and analysis should be conducted.

3.7.2 Fatigue Study

Again, both material implants sustain the expected lifetime of a million cycles without significant internal damage.

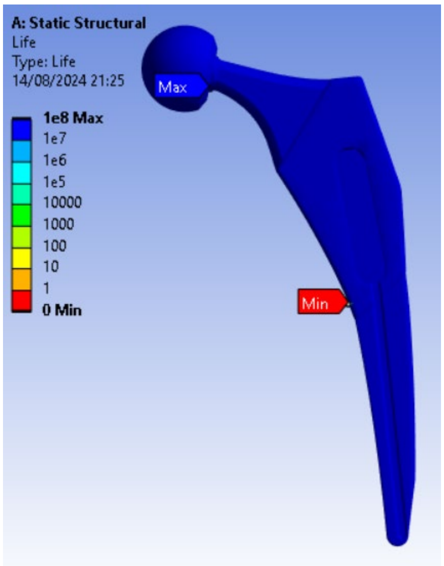


Figure 27: Fatigue life results for Titanium Alloy

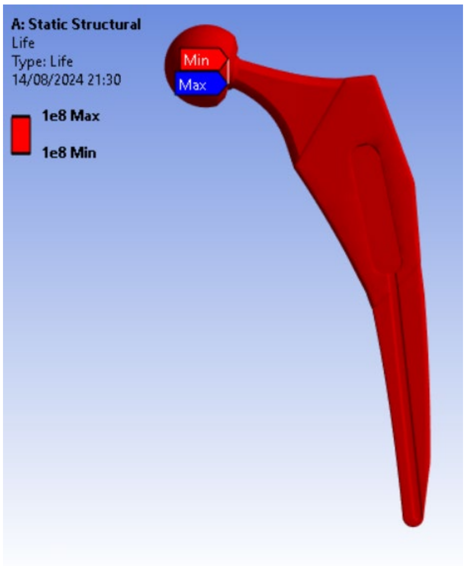


Figure 28: Fatigue life results for Stainless Steel

4 Discussion

4.1 FEA Errors

There are three types of errors that can lead to results that may not fully reflect real-world outcomes, which have been previously mentioned throughout the report:

1. **Modelling Error:** This type of error arises from differences between the features and setup of the finite element (FE) model and the real-life scenario. Examples include incorrect geometry, boundary conditions, or material properties. In this case, it's unlikely that exactly half of the lower stem is fully fixed, as some flex between the bone cement and the stem is expected. Another deformation source could be the loading force is likely to change angles during walking, rather than being constantly downward in the z-plane.

2. **Discretization Error:** This error occurs due to the approximation of the model into finite elements. Mesh refinement is the most effective way to reduce this error, so multiple mesh sizes were investigated to find the most suitable one. Analysing the Jacobian and aspect ratios, which relate to the size and shape of elements, also helps to minimize errors. Mesh analysis indicates that refinement could be further improved through more in-depth mesh optimization.

3. **Numerical Error:** This error arises from the limitations of storing variables during calculations. It can be minimized by using higher precision variables for data storage. This category also includes integration errors from the Gauss integral and rounding errors, which are particularly significant when adding or subtracting very large and very small numbers. However, numerical errors generally have a relatively small impact on the overall results.

4.2 Conclusion

After successfully modifying the original hip implant and conducting mesh analysis on two different implants made from Titanium alloy Ti-6Al-4V and Stainless Steel 316L, several key conclusions were reached:

The revised titanium implant proved to be the superior choice. Although it only achieved three vibrational modes within the desired range, it was really close to being within the suitable range which proves that with further reductions in stiffness and increases in mass this goal could be achieved. Additionally, it passed the static test.

In contrast, the revised stainless steel hip implant failed during static analysis, as the maximum stress exceeded the material's yield stress, making the implant unsuitable for use. But it proved to get closer to the desired 5 models.

For the fatigue analysis, the alternating stress for all implants was below the fatigue limit, indicating that they could endure the required 5 million load cycles.

Additionally, several potential improvements for the finite element (FE) model were identified. Further mesh refinement, particularly focusing on improving the aspect ratio to reduce discretization errors, along with revisiting the fixed support conditions to minimize modelling errors, were recommended as valuable areas for further investigation.

5 References

[1] Michael Meng (27/04/2017). GrabCAD. (n.d.). "Hip Implant Ceramic-on-Ceramic" [CAD file]. Retrieved from: https://grabcad.com/library/hip-implant-ceramic-on-ceramic-1

[2] Niinomi, M. (2008). Mechanical biocompatibilities of titanium alloys for biomedical applications. Journal of the Mechanical Behavior of Biomedical Materials, 1(1), 30-42. https://www.sciencedirect.com/science/article/pii/S1751616107000148

[3] Natural frequency formula taken from DE2 FEA class notes

[4] Gerber means stress equation taken from DE" FEA class notes

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